

Dennis G. Pezan

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EDUCATION

B.S., Operations Research – Finance Specialization

Class of 2029

United States Military Academy (USMA), West Point, NY

Minor in Artificial Intelligence

GPA 3.586

RESEARCH INTERESTS

Reinforcement learning, reward design and misspecification, ensemble methods, predictive modeling under uncertainty, autonomous systems, military operations research

RESEARCH EXPERIENCE

Reward Misspecification in a Simulated Multi-Echelon Resupply Task

Summer 2026

Research Intern, Modeling & Analysis Division, Transformation Decision Analysis Center (TDAC), U.S. Army, Fort Lee, VA

- Engineered and iteratively redesigned the reward function for a reinforcement learning agent operating in the Army's Logistics Battle Command (LBC) simulation
- Diagnosed reward hacking through behavioral auditing, rebuilt the objective, and verified the correction against the original failure modes
- Trained and evaluated the agent in Python (PyTorch) against a new implementation of the LBC environment, analyzing how reward design shaped learned policy behavior

Behavior of Machine Learning Models Under Uncertainty

2025–Present

Undergraduate Researcher, Independent Research (MA289), Department of Mathematical Sciences, USMA

- Conduct applied AI and ML research on model behavior and data-driven decision making
- Developed Python pipelines (PyTorch, scikit-learn) to train and benchmark reinforcement learning and ensemble models; automated experiment tracking and visualization
- Trained and evaluated Decision Tree, Bagging, and Random Forest classifiers on an identical 80/20 split of the Titanic dataset to isolate the effect of ensemble structure
- Computed pairwise tree correlation empirically ($\rho \approx 0.76$ bagging vs. 0.67 random forest) to demonstrate how decorrelation lowers the irreducible variance floor
- Built and deployed an interactive research site presenting the methodology and results

TECHNICAL REPORTS

Pezan, D. G. (2026). *Diagnosing and correcting reward misspecification in a simulated multi-echelon resupply task.*

Technical brief. dennis-pezan.com/rl-reward-paper.html

PRESENTATIONS

Undergraduate Research Presentation, USMA, West Point, NY

Spring 2026

Decision trees and ensemble methods: variance reduction through tree decorrelation

LEADERSHIP EXPERIENCE

Cadet Ambassador

2025–Present

CLD-STEM / Cadet Ambassador Program, USMA

- Represent West Point at advisory boards in Chicago and Los Angeles
- Deliver STEM outreach and mentorship to underrepresented students (50+ hours)

Elected Alderman and Township Supervisor

June 2024

Illinois Premier Boys State

- Elected to municipal and township office; advanced to the Lieutenant Governor primary through competitive public speaking

SERVICE AND OUTREACH

Academic Tutoring 2025–Present
Certified Tutor, Center for Enhanced Performance (CEP), USMA
– Provide tutoring in mathematics, chemistry, and psychology, supporting cadets in mastering core concepts and exam preparation

Technology Refurbishment and Distribution Jan – July 2024
Volunteer Senior Publicist, Recycle & Reboot, Plainfield, IL
– Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

HONORS AND AWARDS

Stokes Writing Fellowship, USMA 2025–Present
Dean’s List, USMA (both semesters) 26-1, 26-2
Norwegian Foot March Badge 2026

TECHNICAL SKILLS AND COURSEWORK

Programming: Python (PyTorch, scikit-learn, pandas, NumPy, matplotlib), Java (Spring Boot), SQL

Tools: Git and GitHub, Vercel, Microsoft Fabric, Kafka, REST APIs, Excel

Coursework: MA289 (Machine Learning), MA371 (Linear Algebra)

Languages: Spanish (fluent)

ADDITIONAL EXPERIENCE

Department Manager, McDonald’s Sept 2022 – July 2024
Plainfield, IL
– Managed weekly scheduling for 80+ employees with 95% on-time accuracy; oversaw \$20,000+ in weekly supply procurement and operations
– Promoted as the youngest department manager in the region at age 16; sustained 90%+ customer satisfaction

Intern, GrayStone AI Solutions June – July 2024
Remote
– Built Python and Excel tools supporting financial planning and analysis workflows